

Corning® SMF-28® Ultra Optical Fiber

Product Information

CORNING



Corning® SMF-28® Ultra single-mode optical fibers combine industry-leading attenuation, improved macrobend performance, and standard 9.2 μm mode field diameter. These full-spectrum fibers are designed for carrier and data center applications and are backward compatible with the installed base of legacy single-mode fibers. They have bend performance that exceeds Recommendation ITU-T G.657.A1 and are fully compliant with ITU-T G.652.D. SMF-28 Ultra fiber is available in a traditional 242 μm diameter as well as a 200 μm diameter option with reduced coating for smaller, lighter, high-fiber-count cables.

Optical Specifications

Maximum Attenuation

Wavelength (nm)	Maximum Value* (dB/km)
1310	≤ 0.32
1383**	≤ 0.32
1490	≤ 0.21
1550	≤ 0.18
1625	≤ 0.20

*Alternate attenuation offerings available upon request.

**Attenuation values at this wavelength represent post-hydrogen aging performance.

Attenuation vs. Wavelength

Range (nm)	Ref. λ (nm)	Max. α Difference (dB/km)
1285 – 1330	1310	0.03
1525 – 1575	1550	0.02

The attenuation in a given wavelength range does not exceed the attenuation of the reference wavelength (λ) by more than the value α .

Macrobend Loss

Mandrel Radius (mm)	Number of Turns	Wavelength (nm)	Induced Attenuation* (dB)
10	1	1550	≤ 0.50
10	1	1625	≤ 1.5
15	10	1550	≤ 0.05
15	10	1625	≤ 0.30
30	100	1625	≤ 0.1

*The induced attenuation due to fiber wrapped around a mandrel of a specified radius.

Point Discontinuity

Wavelength (nm)	Point Discontinuity (dB)
1310	≤ 0.05
1550	≤ 0.05

Cable Cutoff Wavelength (λ_{cc})

$\lambda_{cc} \leq 1260 \text{ nm}$

Mode Field Diameter

Wavelength (nm)	Mode Field Diameter (μm)
1310	9.2 ± 0.4
1550	10.4 ± 0.5

Dispersion

Wavelength (nm)	Dispersion Value [ps/(nm·km)]
1550	≤ 18
1625	≤ 23.7

Zero Dispersion Wavelength (λ_0): $1304 \text{ nm} \leq \lambda_0 \leq 1324 \text{ nm}$
Zero Dispersion Slope (S_0): $\leq 0.092 \text{ ps}/(\text{nm}^2 \cdot \text{km})$

Polarization Mode Dispersion (PMD)

	Value (ps/√km)
PMD Link Design Value	$\leq 0.04^*$
Maximum Individual Fiber PMD	≤ 0.1

*Complies with ITU-T G.650-2 Appendix IV, ($m = 20$, $Q = 0.01\%$), August 2015.

The PMD link design value is a term used to describe the PMD of concatenated lengths of fiber (also known as PMD_0). This value represents a statistical upper limit for total link PMD. Individual PMD values may change when fiber is cabled.

ColorPro® Identification Technology

SMF-28 Ultra fibers are available in colored and ringmarked variants, enabled by ColorPro® identification technology. Corning fibers with ColorPro® identification technology deliver better efficiency in cable manufacturing, simplify inventory management, and leverage an enhanced fiber product offering.

How to Order

Contact your sales representative, or call the Optical Fiber Customer Service Department:
Ph: 1-607-248-2000 (U.S./Can.)
+44-1244-525-320 (Europe)
Email: cofic@corning.com
Please specify the fiber type, attenuation, and quantity when ordering.



Dimensional Specifications

Glass Geometry

Fiber Curl	≥ 4.0 m radius of curvature
Cladding Diameter	125.0 ± 0.7 μm
Core-Clad Concentricity	≤ 0.5 μm
Cladding Non-Circularity	$\leq 0.7\%$

Coating Geometry

	Standard	Reduced Diameter
Coating Diameter	242 ± 5 μm	200 ± 5 μm
Coating-Cladding Concentricity	< 12 μm	≤ 10 μm

Environmental Specifications

Environmental Test	Test Condition	Induced Attenuation 1310 nm, 1550 nm, and 1625 nm (dB/km)
Temperature Dependence	-60°C to $+85^{\circ}\text{C}^*$	≤ 0.05
Temperature Humidity Cycling	-10°C to $+85^{\circ}\text{C}$ up to 98% RH	≤ 0.05
Water Immersion	$23^{\circ}\text{C} \pm 2^{\circ}\text{C}$	≤ 0.05
Heat Aging	$85^{\circ}\text{C} \pm 2^{\circ}\text{C}$	≤ 0.05
Damp Heat	85°C at 85% RH	≤ 0.05

Operating Temperature Range: -60°C to $+85^{\circ}\text{C}$

*Reference temperature = $+23^{\circ}\text{C}$

Mechanical Specifications

Proof Test

The entire fiber length is subjected to a tensile stress ≥ 100 kpsi (0.69 GPa). Higher proof test levels are available.

Length

Fiber lengths available up to 50.4 km/spool.

Performance Characterizations

Characterized parameters are typical values.

Numerical Aperture	0.14 NA is measured at the one percent power level of a one-dimensional far-field scan at 1310 nm.
Effective Group Index of Refraction (n_{eff})	1310 nm: 1.4676 1550 nm: 1.4682
Fatigue Resistance Parameter (n_d)	20
Coating Strip Force	Dry: Standard, 0.6 lbs. (3 N) Reduced Diameter, 0.5 lbs. (2 N) Wet, 14-day room temperature: Standard, 0.6 lbs. (3 N) Reduced Diameter, 0.5 lbs. (2 N)
Rayleigh Backscatter Coefficient (for 1 ns Pulse Width)	1310 nm: -77 dB 1550 nm: -82 dB